Chapter 1

INTRODUCTION

Chapter 1

1 Introduction

1.1 GENERAL OVERVIEW

An E-healthcare management system is a comprehensive digital platform that integrates various components of healthcare management, including electronic health records (EHRs), patient scheduling, billing and payment processing, telemedicine, data analytics, and more. One of the primary advantages of an E-healthcare management system is the digitization of patient health records. By securely storing and organizing medical information in electronic format, healthcare professionals can access patient data in real-time, enabling them to make more informed and timely decisions. This leads to improved patient care, reduced medical errors, and enhanced coordination among healthcare providers.

1.2 OBJECTIVES OF PROJECT

- To optimize bed occupation.
- To improve the use of operating theaters, avoiding the cancellation of operations.
- To know how emergencies affect to the administration of the hospital departments or services (cancellation of operations etc.).
- To optimize the allocation of human and material resources to wards and shifts.
- To detect the influence of certain diseases in the hospital's services.
- To find clusters of patients.

1.3 EXISTING SYSTEM

In the existing health care system, there is a high chance of miss interpretation of data as well as occurrence of errors. Moreover, it is cumbersome and time consuming. With the increase in volume of patients in the health care institutes, traditional method of management has gone out of phase. As a result of this, an advanced Health Care Management System has been the demand of time.

- **Time Consuming:**

Developing and maintaining a E-Health Care System can be a time-consuming process. Firstly, creating high-quality course content that is engaging and informative requires significant effort and resources.. It involves researching the latest trends and best practices in a specific domain, designing instructional materials.

1.4 PROPOSED SYSTEM

In the proposed system the doctor's patients are brought to one platform will allow patients to be more flexible they can register and search for the doctors basing on the location the list of doctors will be shown and patient can book by selecting the time slots and the doctor will confirm the booking so everything is computerized an done very fast which will save time. The patient can select any particular doctor and view his profile. Also the patient can view the doctor's schedule and look for an appointment according to his convenience.

1.5 PLANNING OF PROJECT

The project schedule starts with selection of appropriate topic of the project. The selected is "E-Health Care System Project". This task was completed in April 2024. The process includes searching on the internet, contact professionals and collect the information about our topic from them. Development of Exam Scheduler includes following steps:

- Understanding the project.
- Submitting synopsis.
- Learning period.
- Analysis of project.
- Strategy of project development.
- Design of back end.
- Coding phase.
- Documentation of project.

Chapter 2

Requirement Analysis

Chapter 2

2 REQUIREMENT ANALYSIS

2.1 ANALYSIS OF PROJECT

Following features are included in our project:

- Identify the target audience:
- Define the skills to be covered:
- Analyze the competition:
- Develop a content strategy:
- Consider accessibility:
- Plan for user feedback:

2.2 PROPOSED SYSTEM INFORMATION

Following features are included in our project:

- User-friendly Interface
- Interactive Content
- Personalized Learning
- Gamification Features
- Accessibility Features

Chapter 3

TOOLS

Chapter 3

3 TOOLS

3.1 HARDWARE REQUIREMENTS

For developing the project:

- Processor : Intel i5
- Min Ram : 10GB
- Min Hard Disk Space : 250GB
- I/O Device : Mouse As Pointing Device & Keyboard To Input Data

3.2 SOFTWARE REQUIREMENTS

For developing the project:

- Operating System : Window10
- Front End : HTML,CSS,JSP
- Back End : MySql
- Languages : Java

Chapter 4

Project Design

Chapter 4

4 PROJECT DESIGN

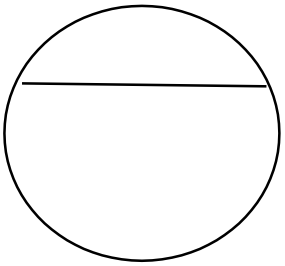
4.1 SYSTEM DESIGN

PROJECT MODULES

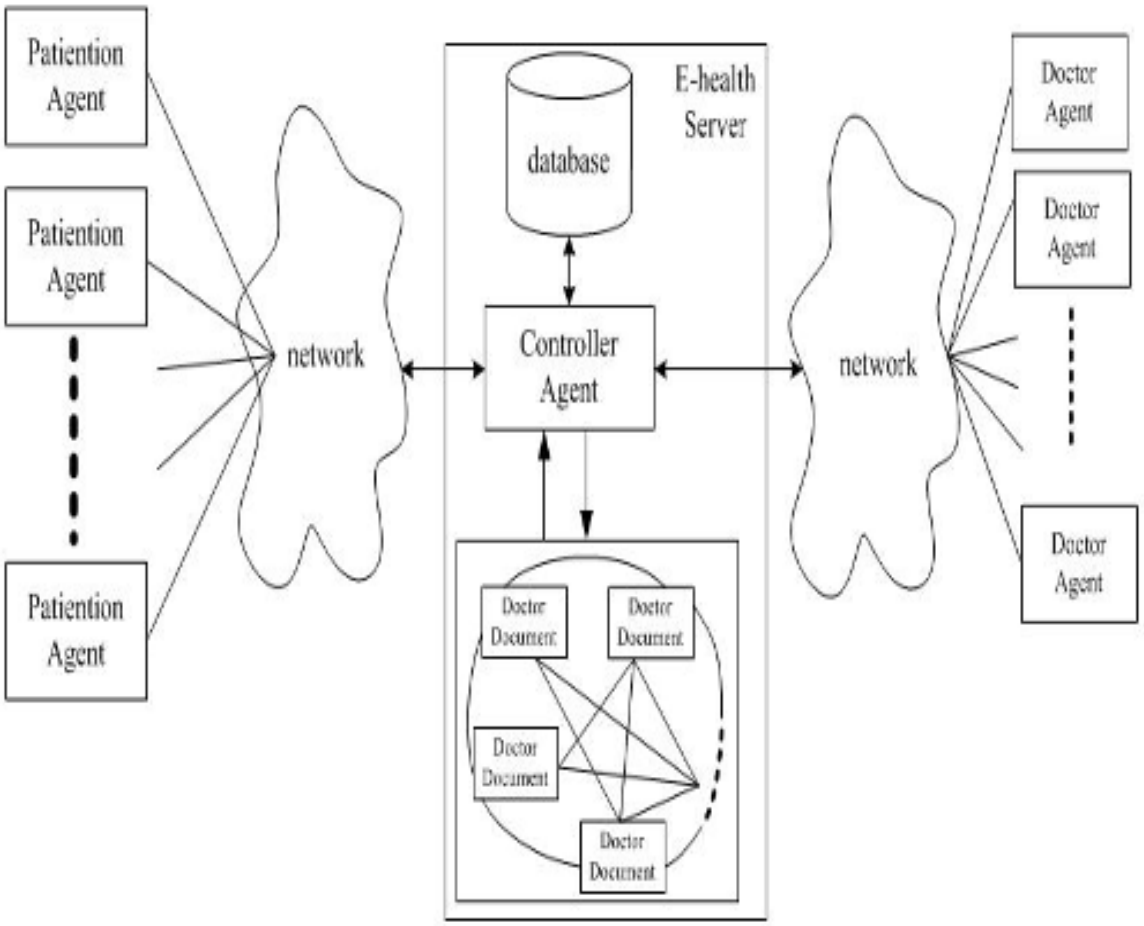
1. System Architecture.
2. Database Architecture module.

- Data Flow Diagrams:

DFD is structured, diagrammatic technique for showing the functions performed by a system and the data flowing into, out of and within it. It was first developed by Larry Constantine as a way of expressing system requirements in a graphical form which led to modular design. It consists of data flows processes, sources, destinations, stores all describes through the use of easily understood symbols. These symbols are:

- External Entity 
- Processes 
- Data flows
- Data stores

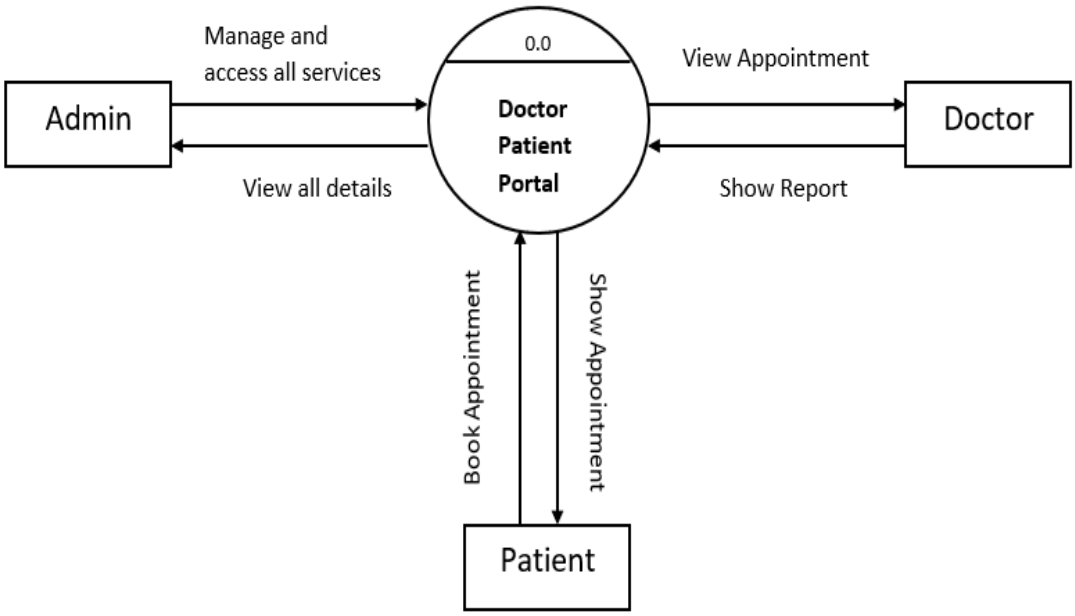
4.2 System Architecture:-



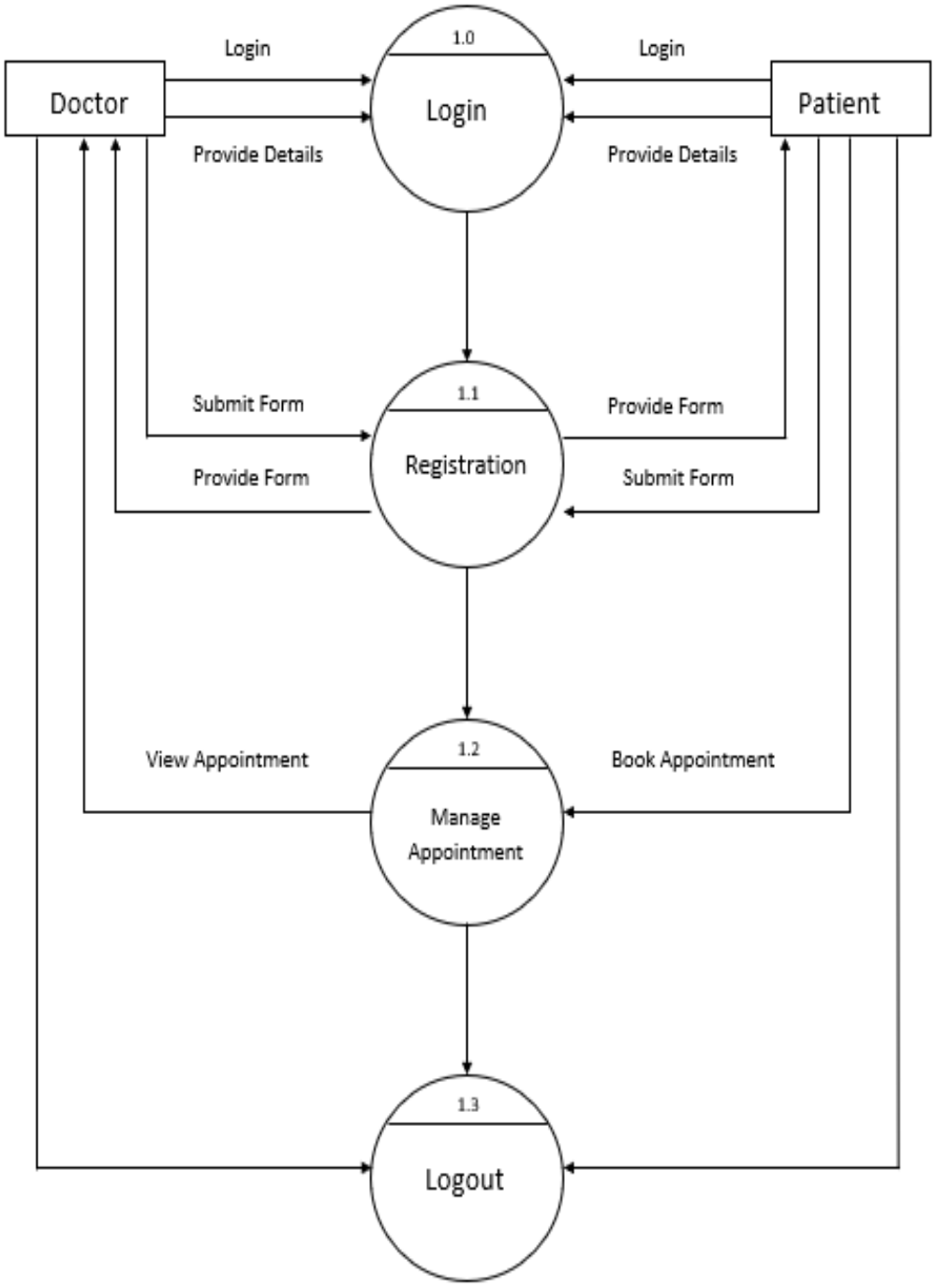
\

4.3 DFD

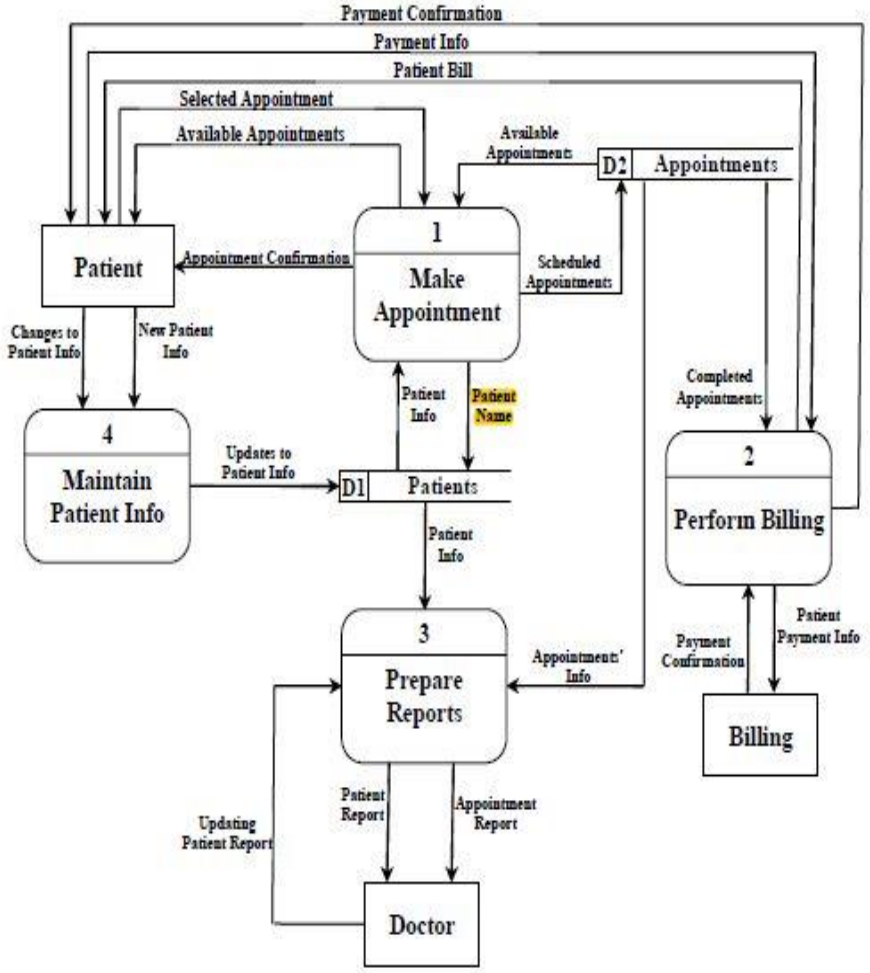
4.3.1 DFD for 0th Level:



4.3.2 DFD for 1st Level:



4.3.3 DFD for 2nd Level:



Chapter 5

Form Design

Request Page:-

```
<%--
   Document   : Patient_Login
   Created on : 14 Feb, 2023, 10:22:21 PM
   Author     : Pratiksha Kesarkar
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<title>JSP Page</title>
<link rel="stylesheet" href="bootstrap.css">
<style type="text/css">
        .jumbotron
        {
width:40%;
margin: auto autoautoauto;
margin-top: 100px;
padding-left: 20px;
padding-right: 20px;

background: whitesmoke;



        }

</style>
```

```
</head>
<body>

<div class="jumbotron">

<center><h2><B>Login</B></h2></center>
<div class="row">
<div class="col-sm-12">
<div class="form-group">
<label>Username</label>
<input type="text"  name="txt_username" class="form-control"
placeholder="Username">
</div>
</div>
</div>

<div class="row">

<div class="col-sm-12">
<div class="form-group">
<label>Password</label>
<input type="password"  name="txt_password" class="form-
control"placeholder="Enter Password" >
<a style="float: right" href="patient_forgot_password.jsp">Forgot password</a>
</div>

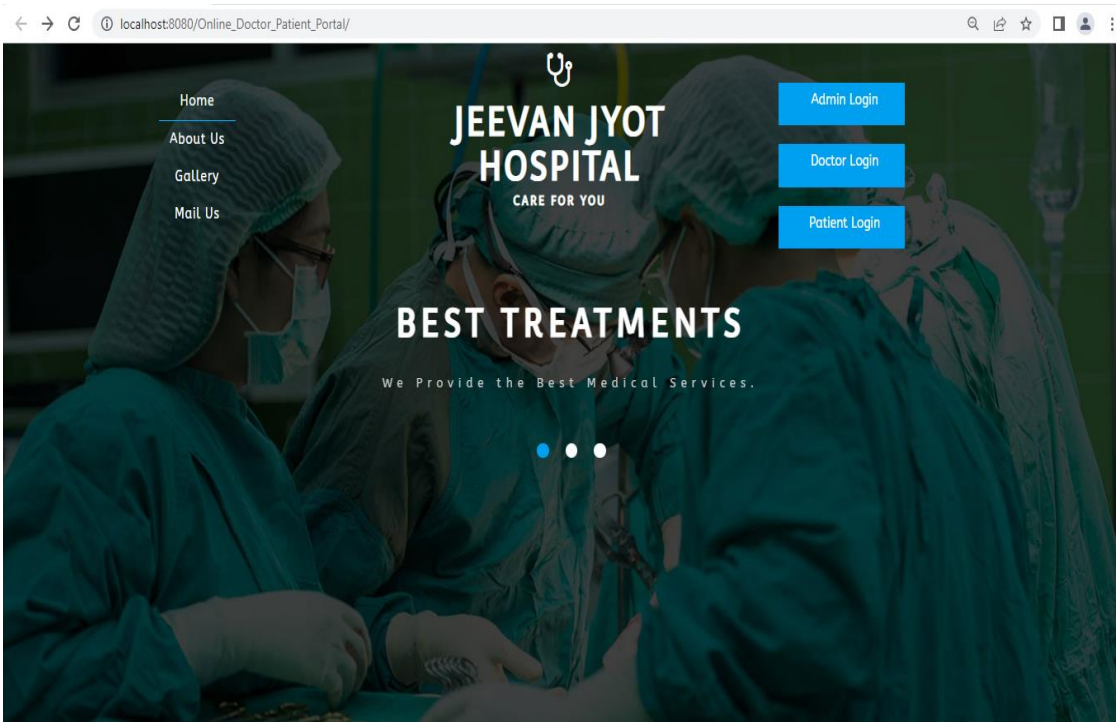
</div>
</div>
<br>
<div class="row">
```

```
<div class="col-sm-4"></div>
<div class="col-sm-4">
<div class="form-group">

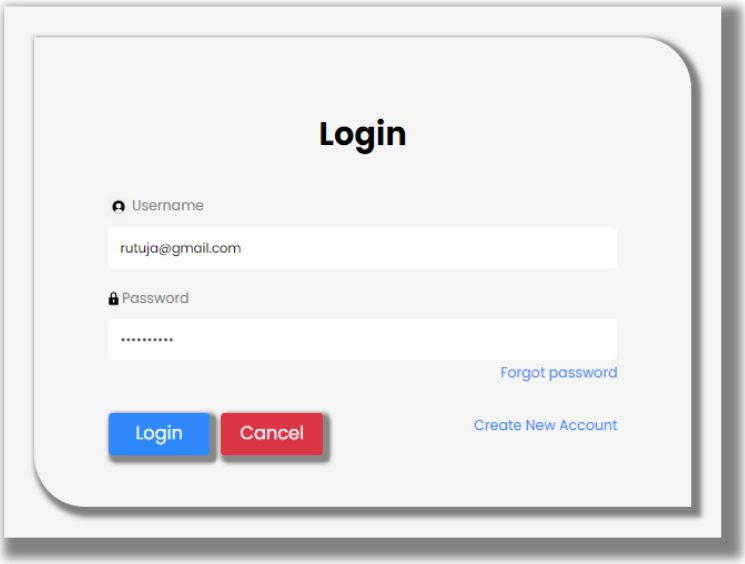
<input type="submit" class="btnbtn-primary" name="btn_save" value="Login">
<input type="submit" class="btnbtn-primary" name="btn_save" value="Cancel">

</div>
<div class="col-sm-4"></div>
</div>
</div>
</div>
</body>
</html>
```

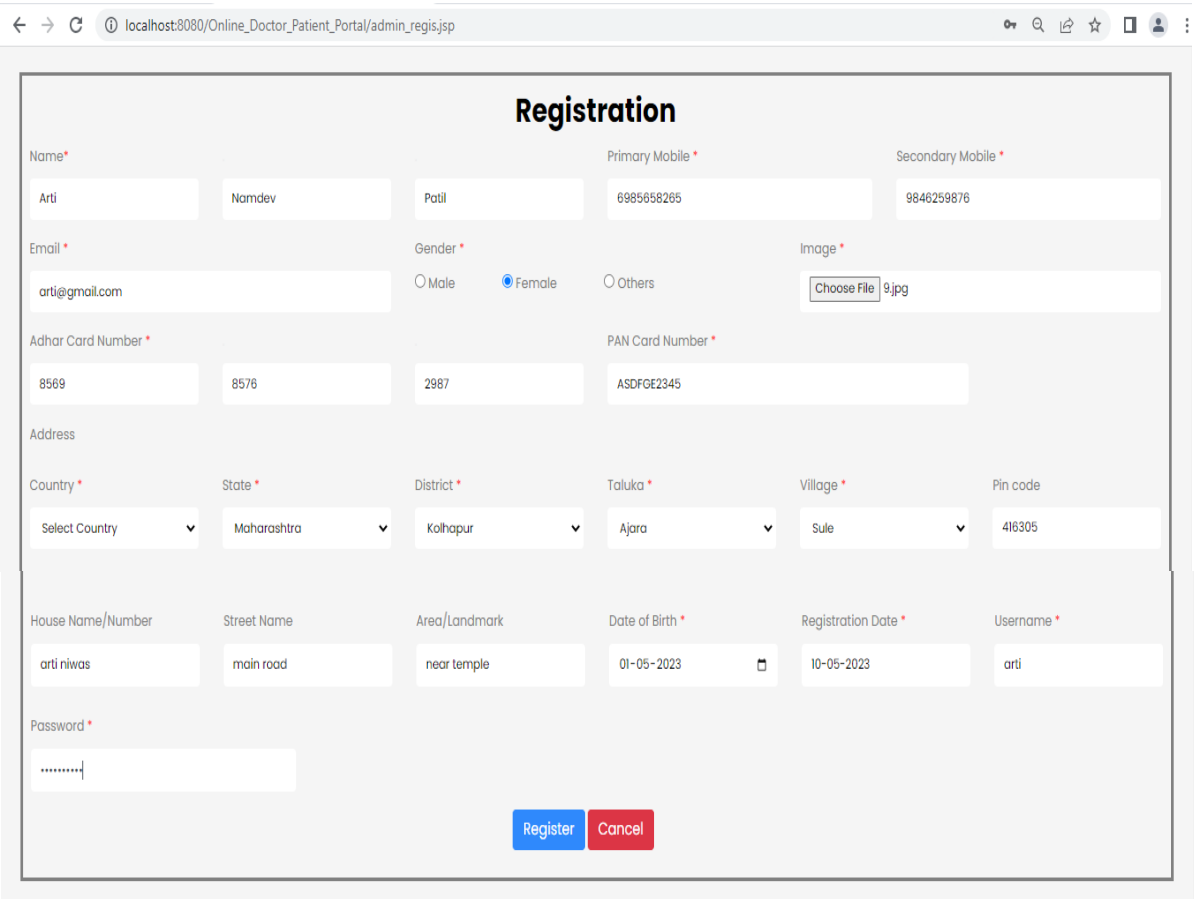
HOME PAGE:-



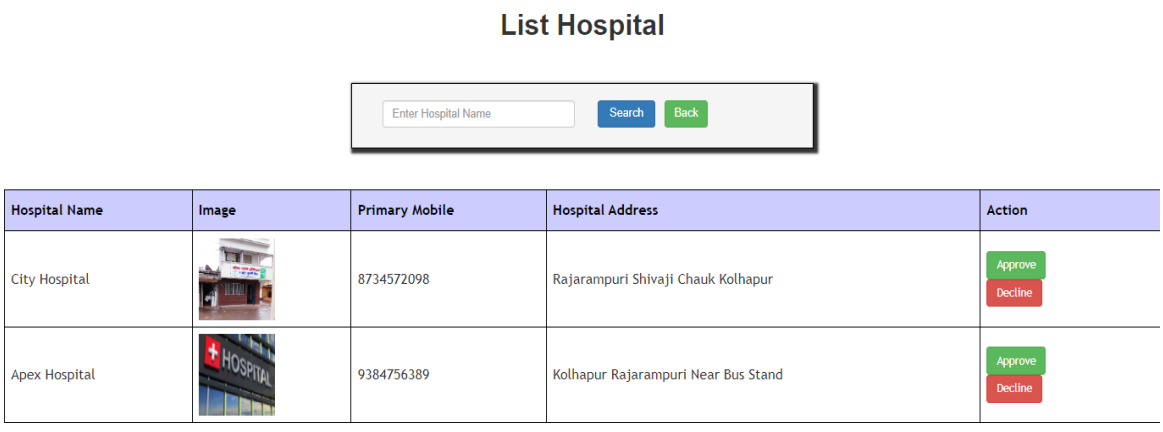
Admin Login:-



Admin Registration:-



List Hospital:-



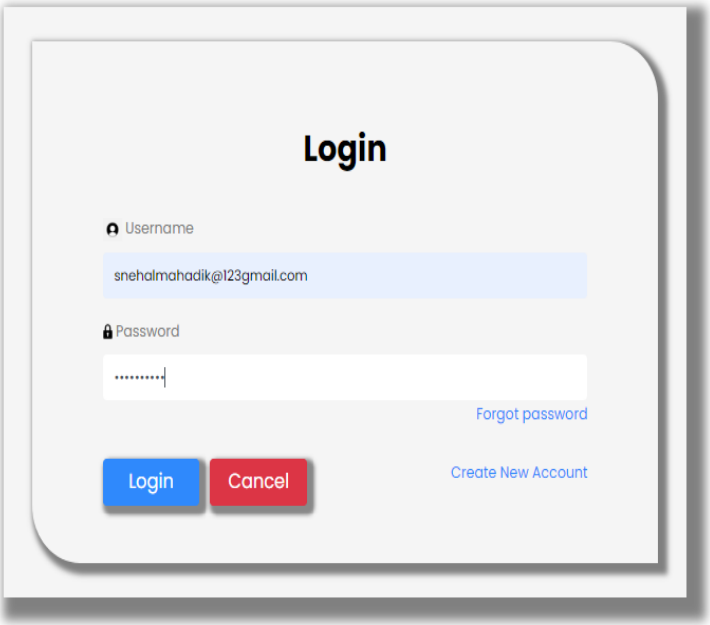
List Doctor:-

List Add Doctor

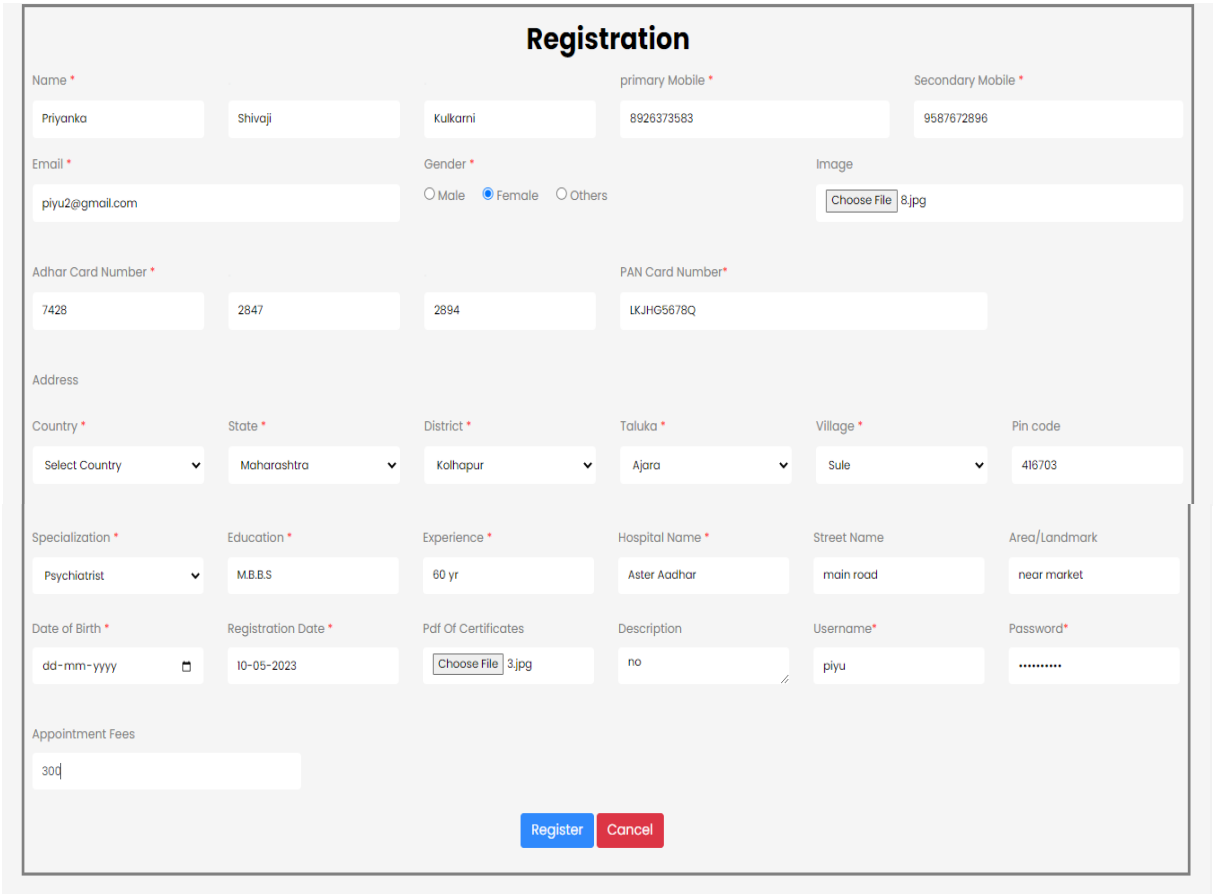
First Name | Middle Name | Last Name | Search | Back

Doctor Name	Primary Mobile	Email	Gender	Education	Experience	Specialization	Image	Hospital Name	Date Of Birth	Registration Date	Action
Shubhangi Anil Chavan	9876893756	shubhangi123@gmail.com	Female	MD	25 yr	Select		City Hospital	2002-05-24	04-05-2023	Approve Decline
Saniya Sanjay Katkar	9758628044	saniya12@gmail.com	Female	MD	15 yr	Select		Apex Hospital	2002-03-12	05-05-2023	Approve Decline

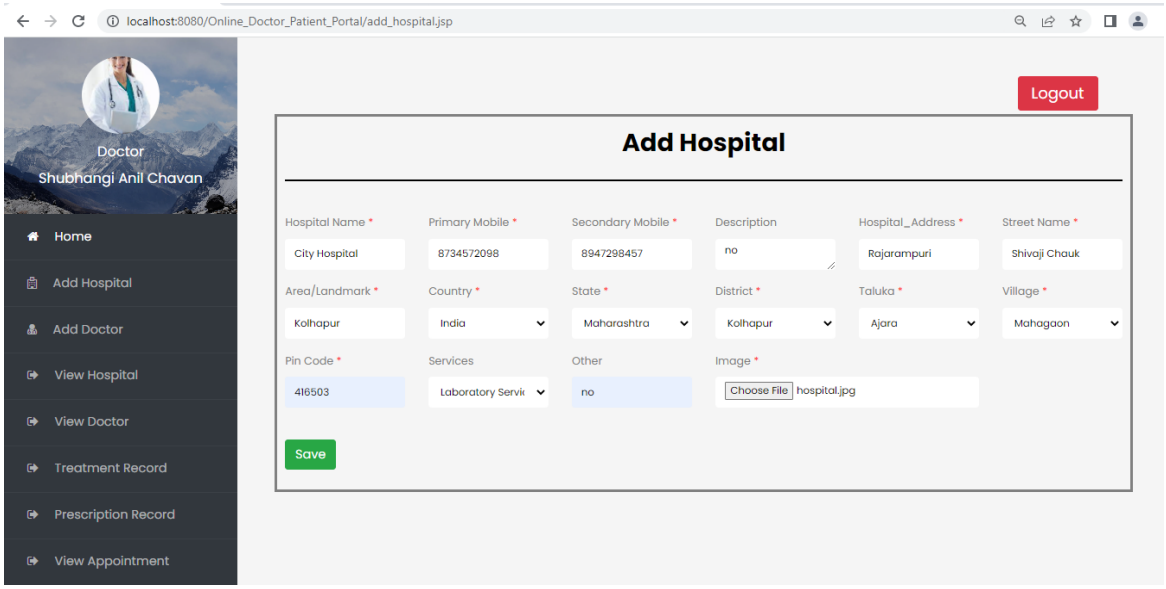
Doctor Login:-



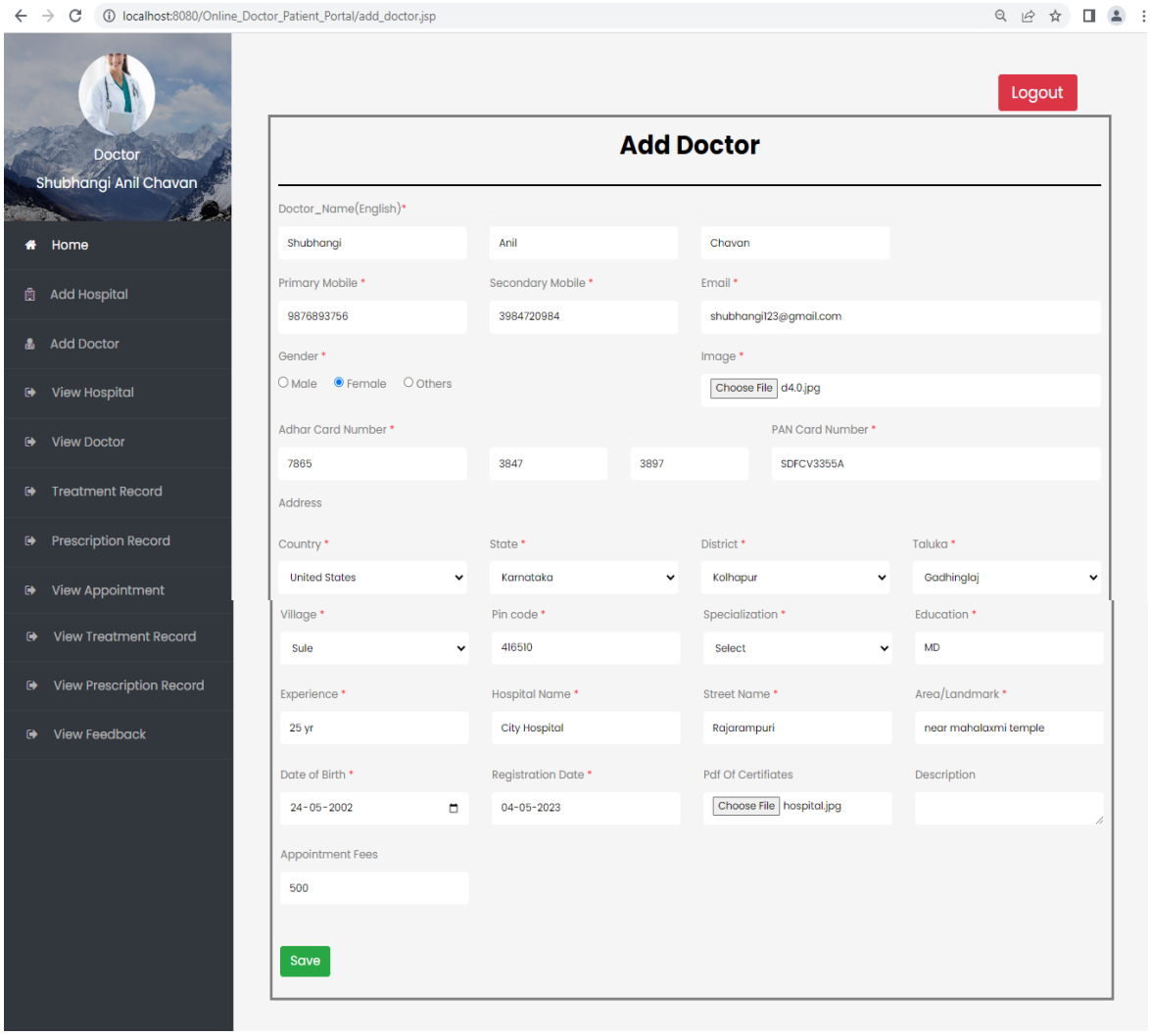
Doctor Registration:-



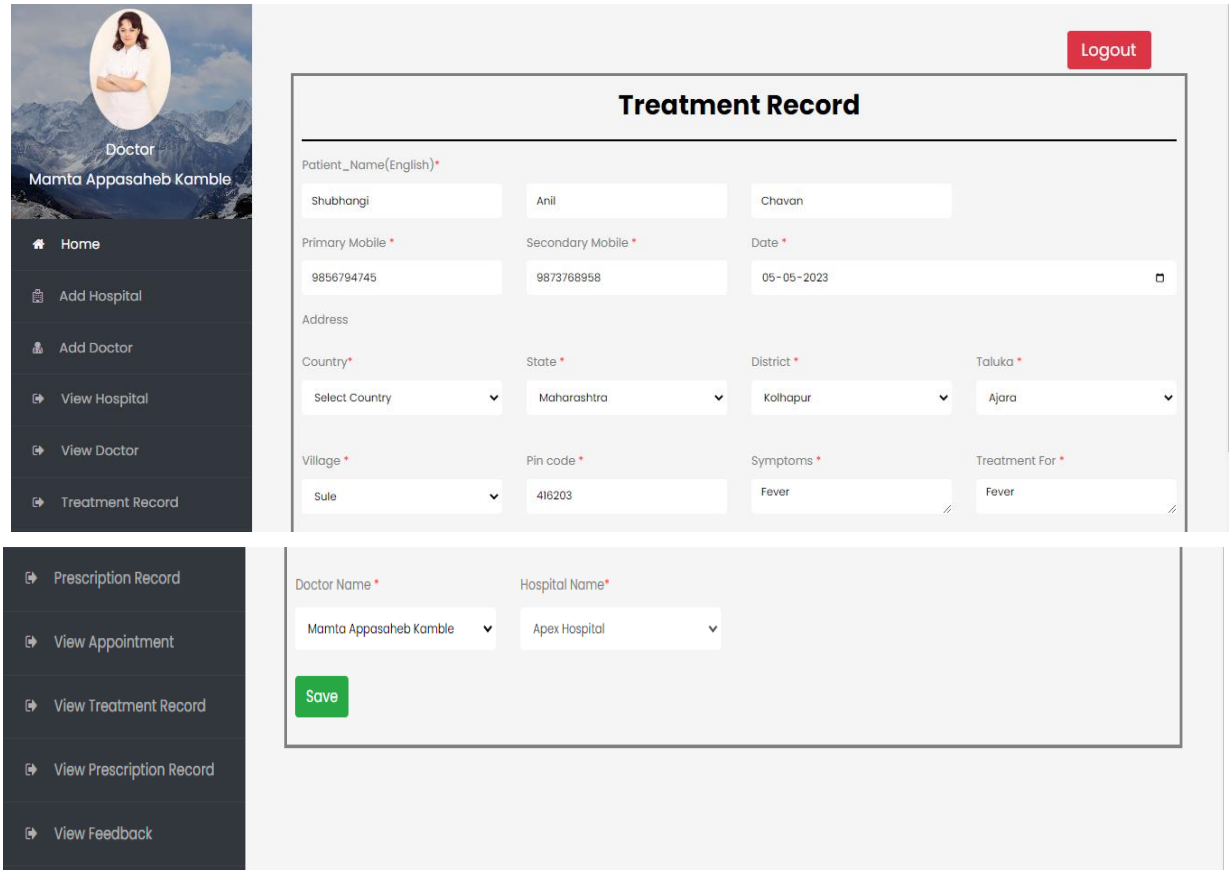
Add Hospital:-



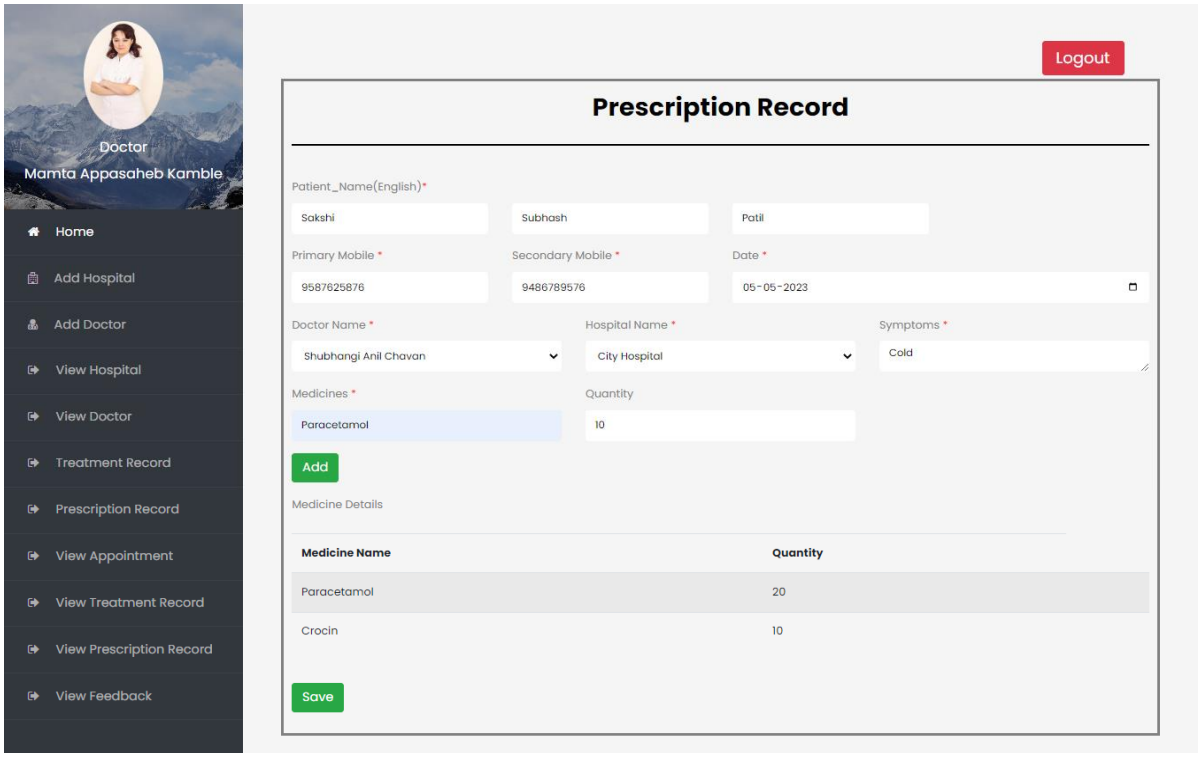
Add Doctor:-



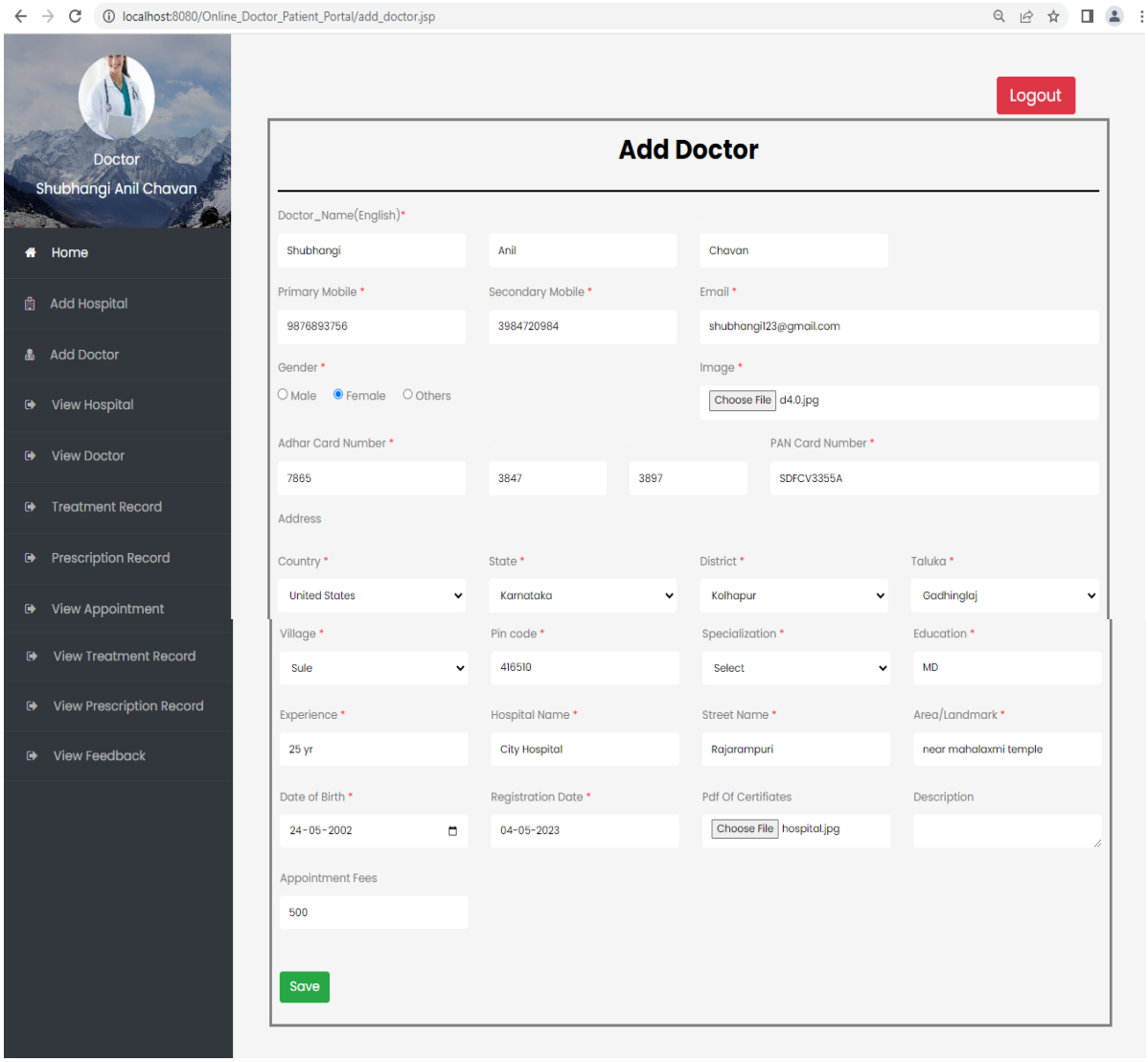
Treatment Record:-



Prescription Record:-



Add Doctor:-


localhost:8080/Online_Doctor_Patient_Portal/add_doctor.jsp
Doctor
Shubhangi Anil Chavan
Home
Add Hospital
Add Doctor
View Hospital
View Doctor
Treatment Record
Prescription Record
View Appointment
View Treatment Record
View Prescription Record
View Feedback
Logout
Add Doctor
Doctor_Name(English)*
Shubhangi
Anil
Chavan
Primary Mobile *
9876893756
Secondary Mobile *
3984720984
Email *
shubhangil23@gmail.com
Gender *
Male
Female
Others
Image *
Choose File d4.0.jpg
Adhar Card Number *
7865
3847
3897
PAN Card Number *
SDFCV3355A
Address
Country *
United States
State *
Karnataka
District *
Kolhapur
Taluka *
Gadhinglaj
Village *
Sule
Pin code *
416510
Specialization *
Select
Education *
MD
Experience *
25 yr
Hospital Name *
City Hospital
Street Name *
Rajarampuri
Area/Landmark *
near mahalaxmi temple
Date of Birth *
24-05-2002
Registration Date *
04-05-2023
Pdf Of Certifiates
Choose File hospital.jpg
Description
Appointment Fees
500
Save

View Appointment:-

List Book Appointment

Enter Doctor Name | Search | Back

Patient Name(English)	Primary Mobile	Email	Gender	Age	DOB	Appointment Date	Appointment Time	Doctor Name	Hospital Name	Action
Pooja Rajendra Kore	9867896738	pooja12@gmail.com	Female	22	2023-05-01	2023-05-05	15:40:00	Mamta Appasaheb Kamble	Apex Hospital	Approve Decline
sushmita Anil Kurale	9843768947	sushmita@gmail.com	Female	22	2023-05-01	2023-05-05	14:49:00	Mamta Appasaheb Kamble	Apex Hospital	Approve Decline

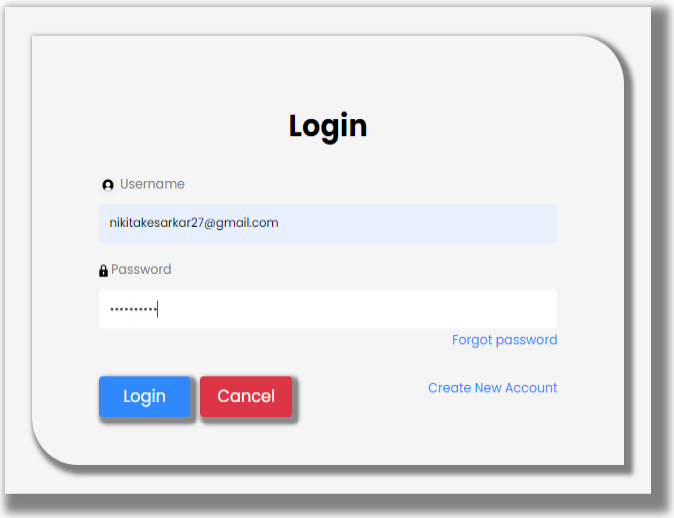
List Hospital:-

List Hospital

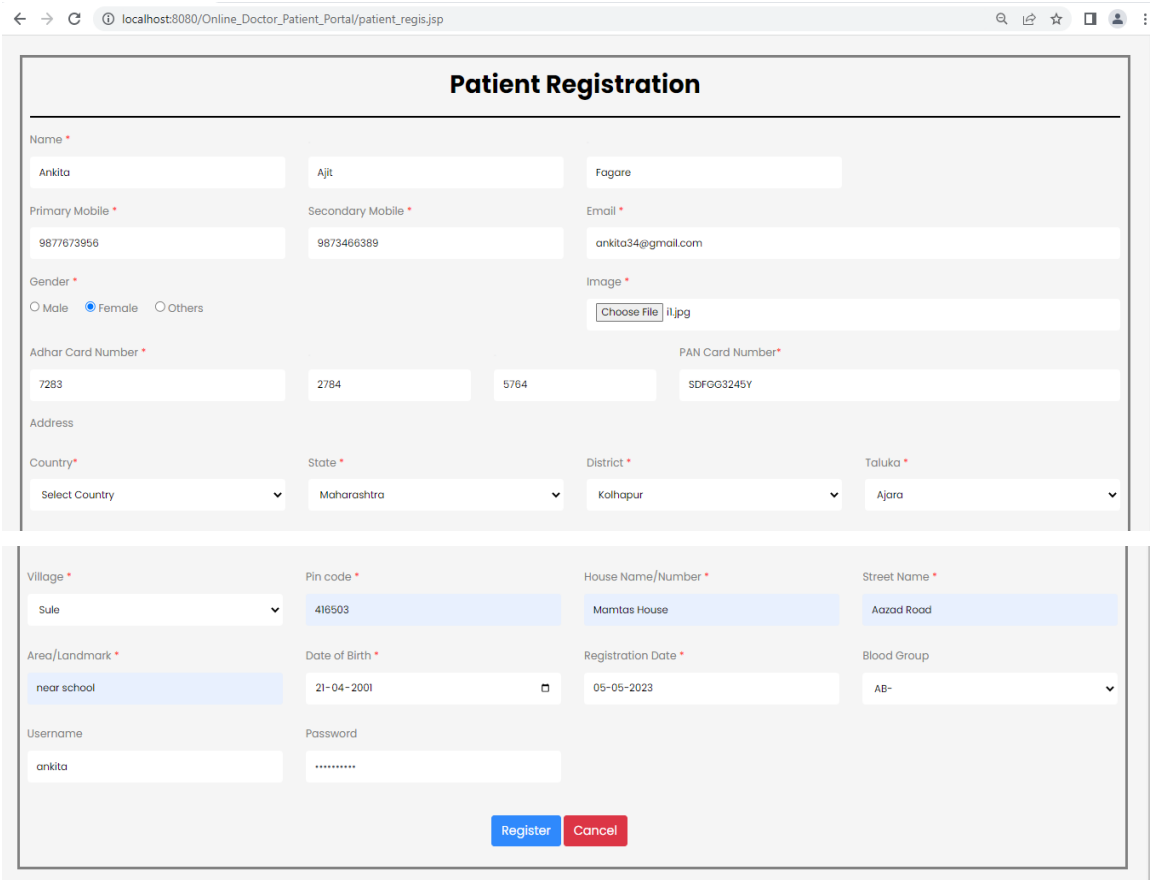
Back

Hospital Name	Image	Primary Mobile	Hospital Address	Status	Action
Apex Hospital		9384756389	Kolhapur Rajarampuri Near Bus Stand	pending	Edit
Sanjivani Hospital		9568728956	Gadhinglaj Main Road near school	pending	Edit

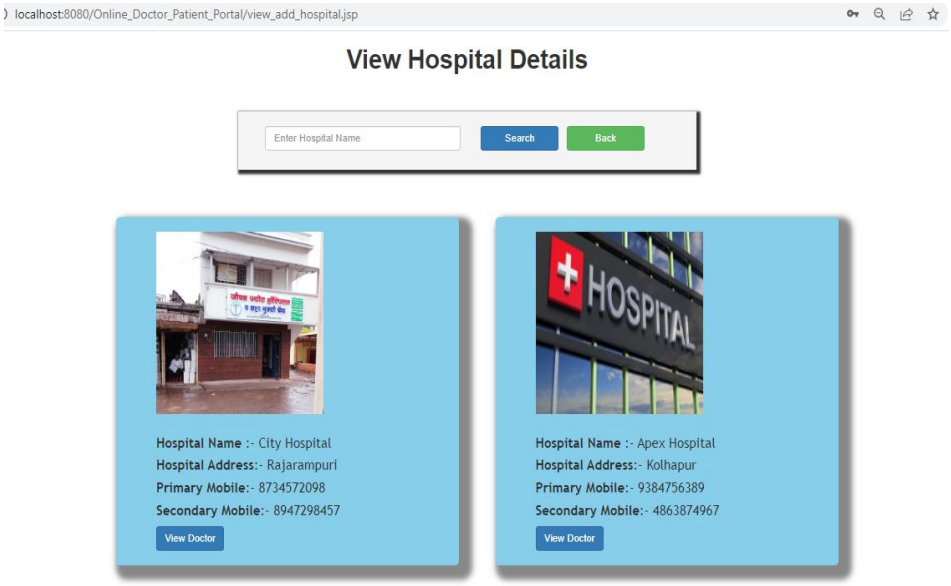
Patient Login:-



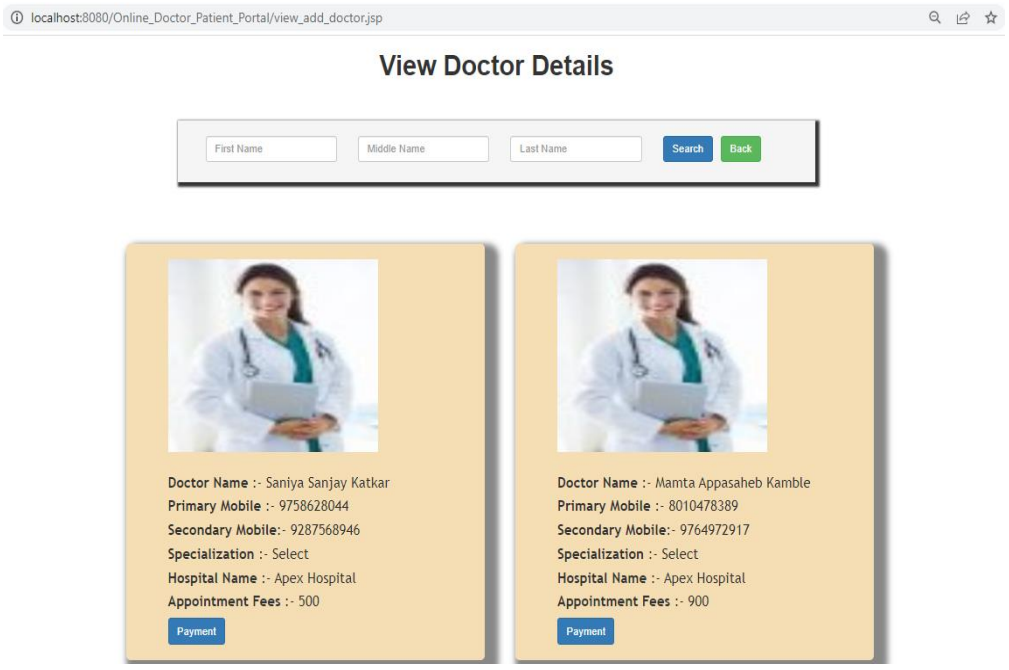
Patient Registration:-



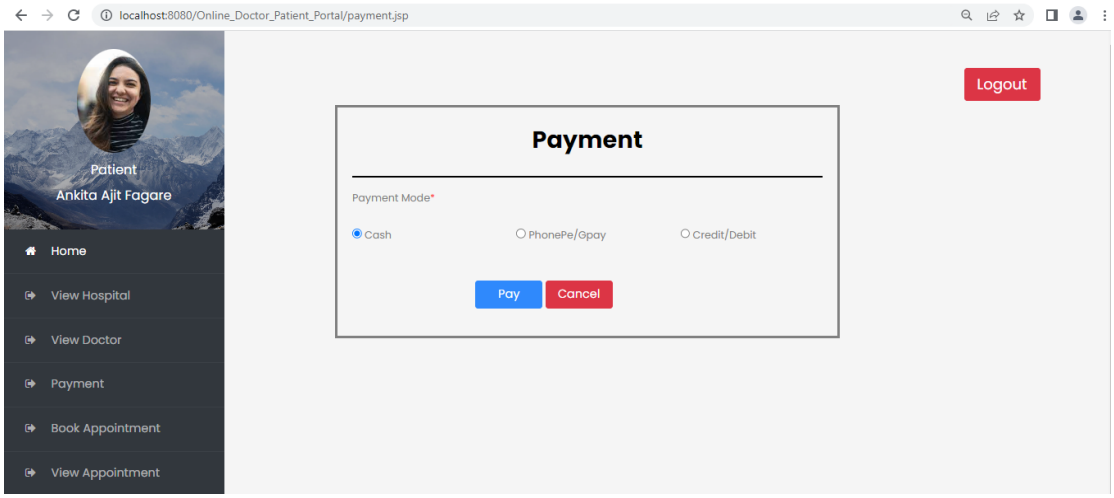
View Hospital:-

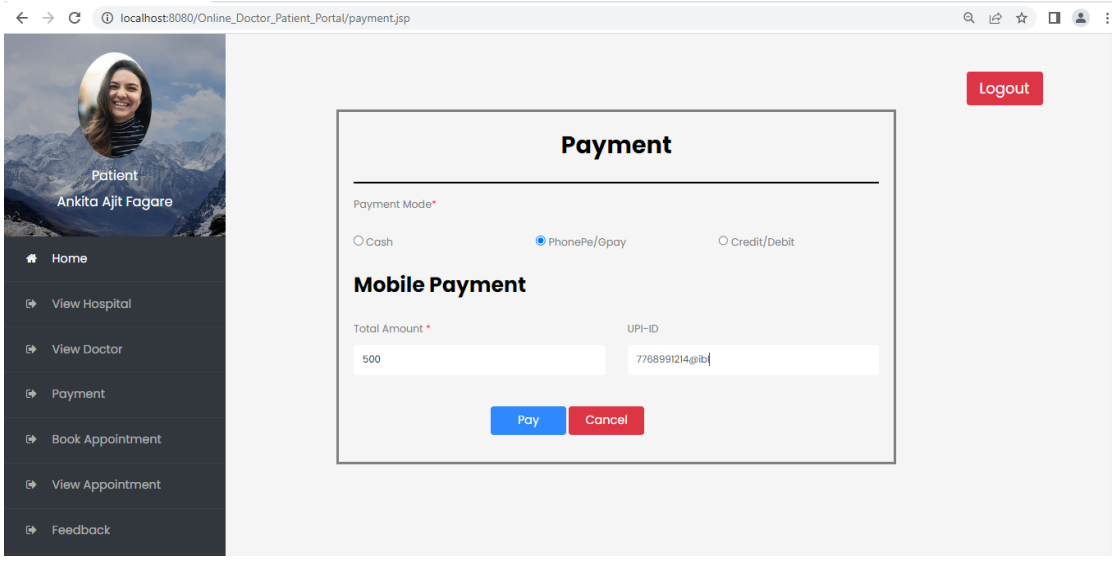


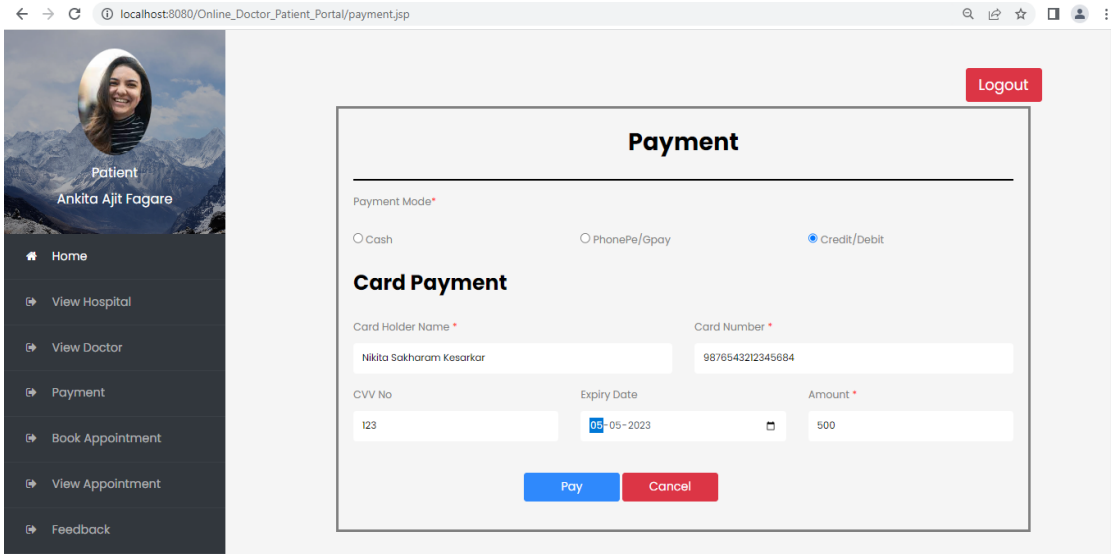
View Doctor:-



Payment:-


localhost:8080/Online_Doctor_Patient_Portal/payment.jsp
Patient
Ankita Ajit Fagare
Home
View Hospital
View Doctor
Payment
Book Appointment
View Appointment
Logout
Payment
Payment Mode*
Cash
PhonePe/Gpay
Credit/Debit
Pay
Cancel


localhost:8080/Online_Doctor_Patient_Portal/payment.jsp
Patient
Ankita Ajit Fagare
Home
View Hospital
View Doctor
Payment
Book Appointment
View Appointment
Feedback
Logout
Payment
Payment Mode*
Cash
PhonePe/Gpay
Credit/Debit
Mobile Payment
Total Amount *
500
UPI-ID
7768991214@ibl
Pay
Cancel


localhost:8080/Online_Doctor_Patient_Portal/payment.jsp
Patient
Ankita Ajit Fagare
Home
View Hospital
View Doctor
Payment
Book Appointment
View Appointment
Feedback
Logout
Payment
Payment Mode*
Cash
PhonePe/Gpay
Credit/Debit
Card Payment
Card Holder Name *
Nikita Sakharam Kesarkar
Card Number *
9876543212345684
CVV No
123
Expiry Date
05-05-2023
Amount *
500
Pay
Cancel

Chapter 7

Validation & Testing

Chapter 7

7 VALIDATIONS AND TESTING

7.1 Testing Levels

The four levels that the analyst uses for the quality assurance are:

1. TESTING
2. VERIFICATION
3. VALIDATION
4. CERTIFICATION

Quality assurance is review of the software product and relates documents for completeness, reliability and maintainability. It includes assurance that the system needs the specification and requirements for its indented use and performance.

System testing is expensive but critical process that can take as fifty percent of the budget for program development. The common view of testing held by user that it is performed to prove that there are no errors in program. However, as indicated alone cannot prove that software is free and clear of errors.

Therefore, the most useful and practical approach s with the understanding that testing is process of executing a program with the explicit intention of finding errors, that is program fails. The tester, who can be analyst, programmer of specialist, trained in software. Testing is actually trying to make a program fail safe. A successful test than is one that find an errors. Analyst knows that an effective program does not guarantee the system reliability. Reliability is the design issue. Therefore reliability must be designed in to system.

Validation prefers the process using software in live environment in order to find errors. The feedback from validation phase generally produces changes in software to deal with errors and failures and may still need changes.

7.2 Advantages of Proposed System

- It is an online system that allows patients to easily book their appointment at a particular hospital related to their health issue.
- Patient does not have to wait in queue to visit the doctor.
- This system saves time of both doctor and patient.
- Doctor can view patient medical history whenever required.
- Patient does not have to carry medical prescription whenever he visits the doctor.

Chapter 8

Future Work

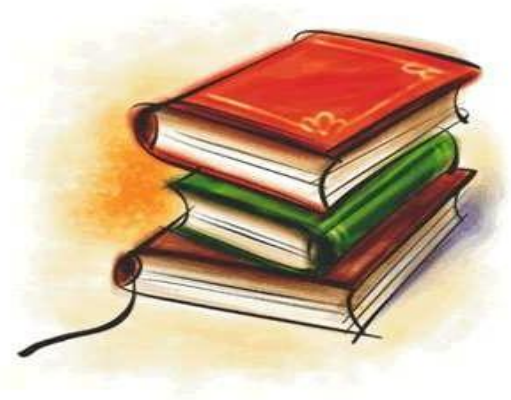
Future Work

In the future, it is expected that the e-health technology will bring revolution in healthcare sector and it needs continuous improvement in developing information and communication technologies. Healthcare professionals will prefer automated systems as patients can also access the e-health data from healthcare service provider. In modern healthcare, it is anticipated that e-health will become key element in the infrastructure and healthcare organizations will need to develop the policies for use of e-health data security. Many hospitals are using biometric system which helps in keeping security of patients' data. In coming days, it will be commonly used in the hospitals for easy and secure access to patients' information. Training of employees and awareness among patients to use e-health technology will be demanding for the growth of e-health in developing countries.

CONCLUSION

In conclusion, This E- health care management system is a web-based application that assists in management of staffs, doctors and patients in easy, comfortable and effective service. Here, the concept of data mining plays vital role to develop an effective health care management system. The proposed application aims to create a friendly working environment for any health care centers and to overcome the drawbacks in existing system of health care management. This system is very reliable and flexible from all aspects, so new features and modules can be easily integrated into the system in future.

Bibliography



Websites

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- www.openai.com
- www.github.com
- www.codewithc.com/e-health-care-management-system-java-project/